

Zero-Knowledge Variant Proof Circuit

Input (Private):

- Patient genotype variants[1..N]
- Query variant (position, allele)

Computation:

1. Compare each variant with query
2. Accumulate matches
3. Check if count > 0

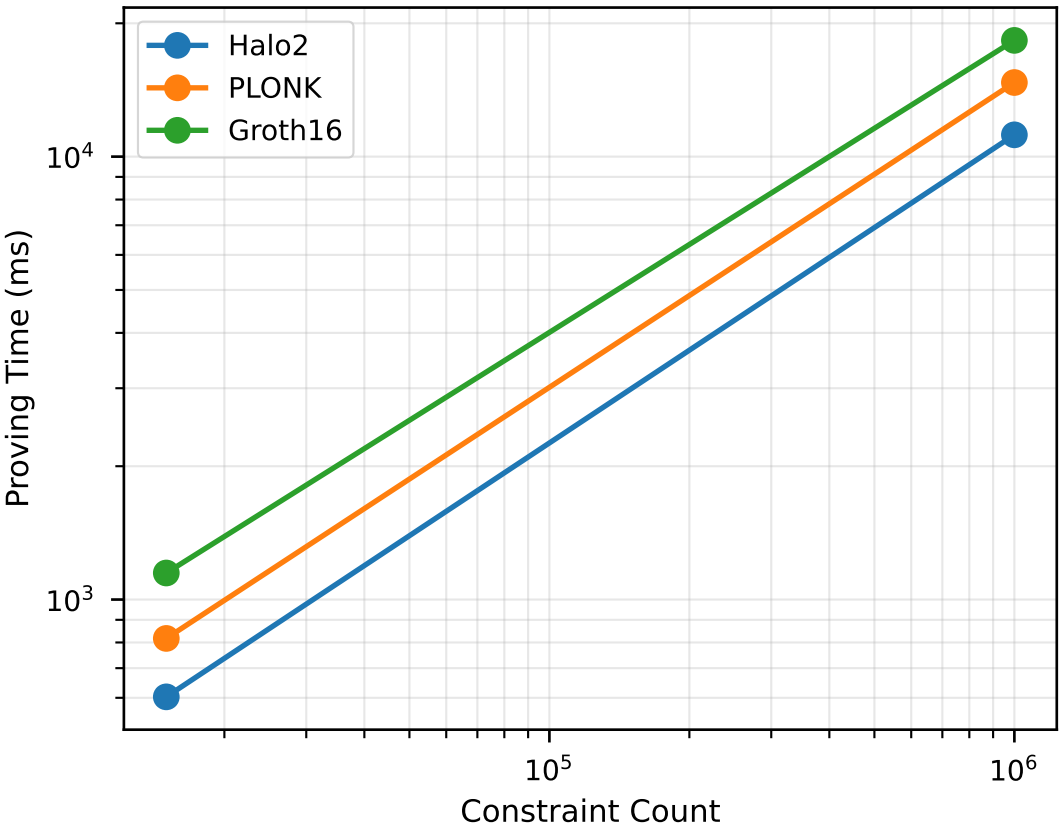
Output (Public):

- Boolean: hasVariant {True, False}
- ZK Proof π (cryptographic proof)

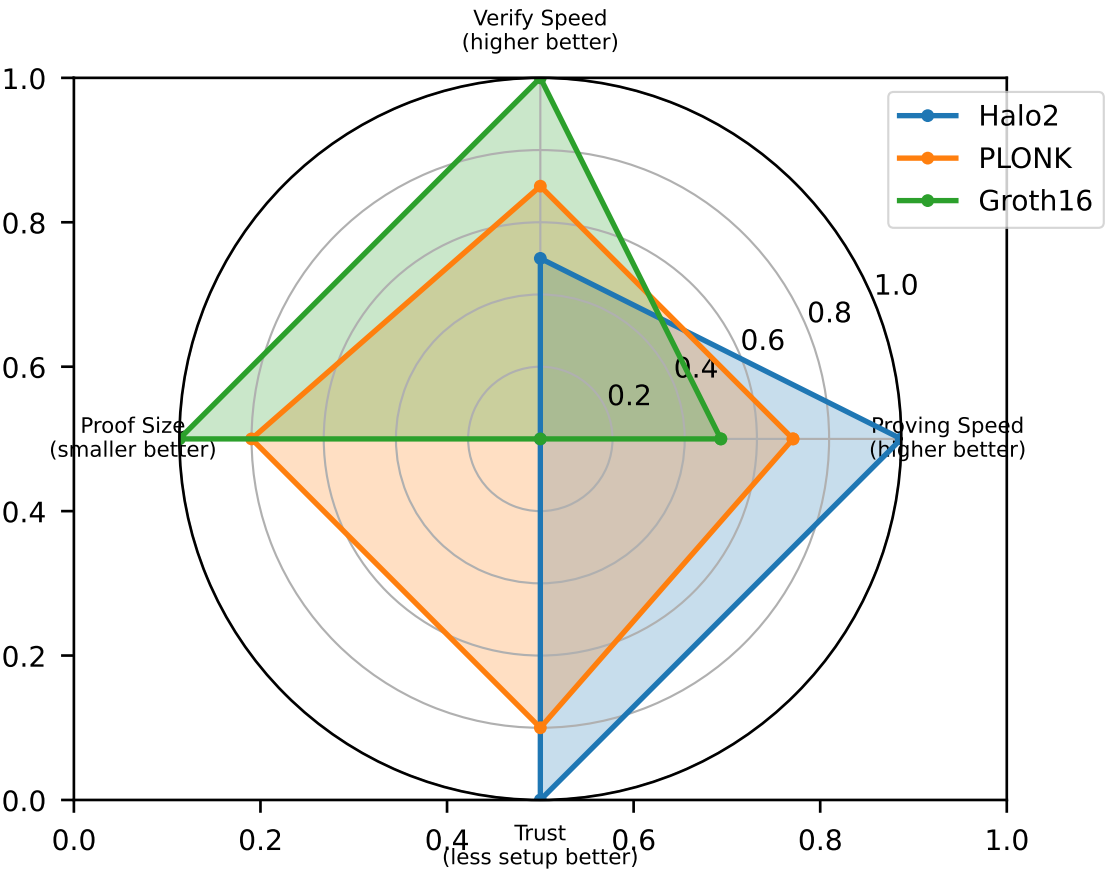
Constraints: 15,234

Privacy: Genotype remains secret, only boolean revealed

B. Proving Time vs Circuit Complexity



D. Backend Comparison (Normalized Metrics)



C. Memory Usage Scaling

